

◆ IAM SESSION 3 — HANDS-ON LABS

AWS Identity & Access Management (IAM)

◆ LAB 1 — CREATE YOUR FIRST IAM USER

Objective

Learn how to create a secure IAM user in AWS.

Tasks

Step 1 — Open IAM

1. Sign in to AWS Console
 2. Search for **IAM**
 3. Open the IAM Dashboard
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Step 2 — Create User

1. Click **Users**
2. Click **Create User**
3. Username:

`student-admin`

4. Enable:
 - AWS Management Console access
 5. Set:
 - Custom password
-

Step 3 — Attach Permissions

Attach policy:

`AdministratorAccess`

Step 4 — Review & Create

Click:

Create User

◆ Expected Outcome

- ✓ IAM user created
 - ✓ Console login enabled
 - ✓ User has administrator permissions
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◆ LAB 2 — ENABLE MFA

Objective

Secure an IAM account using Multi-Factor Authentication.

Tasks

Step 1 — Open User

1. IAM → Users
 2. Select your user
-

Step 2 — Configure MFA

1. Click:

Enable MFA

2. Choose:

Authenticator App

3. Use:
 - Google Authenticator
 - Microsoft Authenticator
 - Authy
-

Step 3 — Scan QR Code

1. Open authenticator app
2. Scan QR code
3. Enter generated codes

◆ Expected Outcome

- ✓ MFA enabled
- ✓ Account protected with device verification

◆ LAB 3 — CREATE IAM GROUP

Objective

Understand permission management using groups.

Tasks

Step 1 — Open Groups

IAM → User Groups

Step 2 — Create Group

Group Name:

CloudAdmins

Step 3 — Attach Policy

Attach:

AdministratorAccess

Step 4 — Add User

Add:

student-admin

◆ Expected Outcome

- ✓ Group created
- ✓ Permissions inherited through group

◆ LAB 4 — CREATE CUSTOM POLICY

Objective

Create least-privilege access for Amazon S3.

Tasks

Step 1 — Open Policies

IAM → Policies → Create Policy

Step 2 — Choose JSON

Paste:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket",
        "s3:GetObject"
      ],
      "Resource": "*"
    }
  ]
}
```

Step 3 — Save Policy

Policy Name:

S3ReadOnlyCustom

◆ Expected Outcome

- ✓ Custom IAM policy created
 - ✓ Read-only S3 permissions configured
-

◆ LAB 5 — ROOT ACCOUNT SECURITY CHECK

Objective

Understand root account protection.

Tasks

Verify:

- MFA enabled on root account
 - No daily use of root account
 - Billing alerts configured
 - Root credentials stored securely
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◆ SECURITY BEST PRACTICES

- ✓ Enable MFA everywhere
 - ✓ Avoid root account usage
 - ✓ Use least privilege permissions
 - ✓ Rotate credentials regularly
 - ✓ Prefer IAM roles over users
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◆ OFFICIAL AWS IAM LAB REFERENCES

[AWS IAM Documentation](#)

[AWS IAM Best Practices](#)

◆ RECOMMENDED VIDEO LAB

[AWS IAM Hands-On Tutorial \(YouTube\)](#)

◆ STUDENT CHALLENGE (BONUS)

Create:

- 1 IAM User
- 1 IAM Group
- 1 Custom Policy
- Enable MFA
- Apply least privilege access

Take screenshots and upload to GitHub repository.